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SECOND-YEAR GROUP RESEARCH PROJECT

Learning Causal Structure and Effects from Observational Data

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Abstract

Learning and discovering Causal structure and effects from observational data is a crucial challenge in statistics and computational biology. In this project, we investigate the use of Linear Non-Gaussian Acyclic Model (LiNGAM) to discover causal relationship in gene expression data. Using single-cell datasets from CausalBench benchmark, we developed a data-processing pipeline for downloading, processing, and analysing large scale gene expression. The data demonstrates non-Gaussian characteristics which motivates the use of LiNGAM to determine directed relations between genes. Our project explores the theoretical principles for identifying causal relationship and the practical challenges when applying causal discovery methods to biological systems.

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1 Introduction

Discovering and understanding causal structures in an important objective in science. It is especially important in computational biology where we are interested in the mechanisms of how genes can influence cellular behavior. While simple correlation can reveal association between genes, they don't necessarily indicate the causal effects between them. Traditionally, causal relations can be determined using controlled experiment; however, this can often be costly and very time-consuming when dealing with large datasets.

As a result, A natural question to ask would be whether we can discover causal structures from observational data. Given observational data record naturally occurring variations without modifications, it can be challenging to determine the causal relations of it since there can be several different causal models that are able to explain the same set of observed correlation. Consequently, we need more modelling assumption in order to determine the unique causal relations.

In this project, we investigate the Linear Non-Gaussian Acyclic Model (LiNGAM), a causal discovery framework under the assumption of Non-Gaussianity of the observational data. After constructing a data-processing pipeline and carrying out analysis on the gene expression data from CausalBench benchmark. We use LiNGAM to discover the causal relationships between genes. The aim of the project is to explore both the theoretical and the practical implementations of LiNGAM and find out any challenges when determining causal structure of real-life datasets.

2 Background

2.1 Causal Structure

When trying to find patterns and causal effects between multiple parameters through analysing data, there could be identifiability issues. For example, if X_1 and X_2 are variables, and we would like to see whether one causes the other, there could be two models:

M_1:

$$\begin{aligned} \epsilon_1 &\sim N(0,1) \\ \epsilon_{X_1} &\sim N(0,1) \\ X_1 &= \epsilon_{X_1} \\ X_2 &= X_1 + \epsilon_1 \end{aligned}$$

Joint Distribution:

$$X_1, X_2 \sim N\left(\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1 & 1\\1 & 2\end{pmatrix}\right)$$

M_2:

$$\begin{aligned} \epsilon_2 &\sim N\left(0, \frac{1}{2}\right) \\ \epsilon_{X_2} &\sim N(0,2) \\ X_1 &= \frac{1}{2}X_2 + \epsilon_2 \\ X_2 &= \epsilon_{X_2} \end{aligned}$$

Joint Distribution:

$$X_1, X_2 \sim N\left(\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1 & 1\\1 & 2\end{pmatrix}\right)$$

We say that for Model 1, X_1 causes X_2, as when X_1 is modified, X_2 will change, and for Model 2, X_2 does not change when X_1 is modified.
The important thing to note here is that although the underlying causal structure is different, the final joint distribution of the two variables X_1 and X_2 are the same for both models, which shows that correlation does not imply causation.

As an example, if we apply a change to one of the parameters, such as setting $X_1 = a$. The notation for this is $do(X_1 = a)$:

$M_1^{do(X_1=a)}$:

$$\begin{aligned} \epsilon_1 &\sim N(0,1) \\ X_1 &= a \\ X_2 &= X_1 + \epsilon_1 \end{aligned}$$

Conditional distribution:

$$X_2 \mid X_1 = a \sim N(a, 1)$$

$M_2^{do(X_1=a)}$:

$$\begin{aligned} \epsilon_2 &\sim N\left(0, \frac{1}{2}\right) \\ X_1 &= a \\ X_2 &\sim N(0,2) \end{aligned}$$

Conditional Distribution:

$$X_2 \mid X_1 = a \sim N(0, 2)$$

In this case, for Model 1, the distribution of X_2 changes depending on the value of a, but for Model 2, the distribution of X_2 is unaffected by the value of a. By fixing certain variables, we can attempt to identify the underlying causal structure between multiple variables.

2.2 Structural Equation Models

Structural Equation Models (SEM) are a tool for describing the process in which the values of variables are generated [1]. Structural equations are used to represent how the value of variables is determined. In general, they are a 4-tuple consisting of[1]:

- endogenous variables (Dependent variables inside the system which could be affected by the model)
- exogenous variables (Independent variables that are unaffected by the system)
- deterministic functions that define the structual equations
- probability distribution of the exogenous variables.

The example in the previous subsection is an example of an SEM.

- X_1 and X_2 are endogenous variables
- ϵ_1, ϵ_2, ϵ_{X_1} and ϵ_{X_2} are exogenous variables
- $f_{X_2}(X_1, \epsilon_1) = X_1 + \epsilon_1$ and $f_{X_1}(X_2, \epsilon_2) = \frac{1}{2}X_2 + \epsilon_2$ are the deterministic functions for M_1 and M_2 respectively.
- $\epsilon_1 \sim N(0,1)$ and $\epsilon_2 \sim N(0, \frac{1}{2})$ are the probability distributions of the exogenous variables.

We can also draw a path diagram to represent the causal structure between the variables.

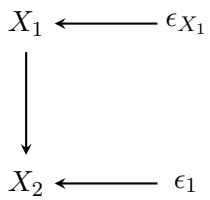


Figure 1 Path diagram for M_1

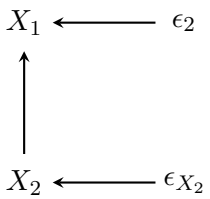


Figure 2 Path diagram for M_2

2.3 Definition of Causation

Intervention on a variable means to hold it constant, for example by applying $M_1^{do(X_1=a)}$. After intervening on the variable, a new distribution for the other endogenous variable would be generated.
We can say that if there are two endogenous variables in a system X_1, X_2 then X_1 causes X_2 if:

$$\begin{aligned} &\exists\, a, b \quad \text{such that} \quad a \neq b : \\ &p(X_2 \mid \text{do}(X_1 = a)) \neq p(X_2 \mid \text{do}(X_1 = b)) \end{aligned}$$

For example, considering Model 1 and assuming it is true, if $a \neq b$, we have:

$$\begin{aligned} & p(X_2 \mid \text{do}(X_1 = a)) \sim N(a, 1) \\ & p(X_2 \mid \text{do}(X_1 = b)) \sim N(b, 1) \\ \Rightarrow \quad & p(X_2 \mid \text{do}(X_1 = a)) \neq p(X_2 \mid \text{do}(X_1 = b)) \end{aligned}$$

Which can be used to show that X_1 causes X_2.

2.4 Causal Effect

A method of quantifying the strength of a causal connection is by calculating the average difference of the following:

$$E(X_2|do(X_1 = d)) - E(X_2|do(X_1 = c))$$

Which is defined as the average causal difference [1]. This evaluates to what extent on average X_2 will change if X_1 is changed from c to d.
Assuming again that Model 1 is true, we have:

$$\begin{aligned} & E(X_2|do(X_1 = d) - E(X_2|do(X_1 = c)) \\ =& E(N(d, 1)) - E(N(c, 1) \\ =& d - c \end{aligned}$$

Which means that the average difference in X_1 is just $d - c$, as the coefficient behind X_1 in the SEM is 1.
Next we can evaluate the average causal difference if X_2 were to be intervened:

$$\begin{aligned} & E(X_1|do(X_2 = d) - E(X_1|do(X_2 = c)) \\ =& E(N(0, 1)) - E(N(0, 1)) \\ =& 0 \end{aligned}$$

Which is to be expected, as X_2 does not cause X_1.

2.5 Impact of Latent confounders on average causal effect

Latent confounders are variables that cause dependence between the exogenonus variables.

$$\begin{aligned} C &\sim N(0, 1) \\ \epsilon_X &\sim N(0, 1) \\ \epsilon_Y &\sim N(0, 1) \\ X &= C + \epsilon_X \\ Y &= C + X + \epsilon_Y \end{aligned}$$

In this case, we can find the true average causal effect by applying intervention, and letting $d = x + 1$ and $c = x$ to find the change when X is increased by 1:

$$\begin{aligned} & E(Y|do(X = x + 1)) - E(Y|do(X = x)) \\ =& E(N(x + 1, 2)) - E(N(x, 2)) \\ =& 1 \end{aligned}$$

However, if we were to find the estimated causal effect using regression:

$$\begin{aligned} Y &= \beta_1 X + \beta_0 \\ \beta_1 &= \frac{Cov(X, Y)}{Var(X)} \\ \beta_1 &= \frac{Cov(X, C + X + \epsilon_Y)}{2} \\ \beta_1 &= \frac{Cov(C + \epsilon_X, C) + Cov(X, X) + Cov(X, \epsilon_Y)}{2} \\ \beta_1 &= \frac{Var(C) + Var(X)}{2} \\ \beta_1 &= \frac{3}{2} \end{aligned}$$

By doing Ordinary Least Square Regression, an incorrect average causal effect would be calculated, as the latent confounder term is not included.

3 Identifying Causal Structures

In practice within the field of statistics, we have a large dataset and as the statistician, want to discover if the data proves the existence of any causal relationships between variables.

Identifiability is the idea that the causal structure can be uniquely estimated given observed data.

We need to understand under which model assumptions identifiability holds. Where model assumptions must represent known background knowledge and constraints on the SEM. Assumptions can be tested to look for violations but cannot be proven true. We start by constructing a basic linear acyclic model:

3.1 A Simple Linear Acyclic Model

Firstly, we must make the following assumptions:

Acyclicity: The causal relations of observed variables are acyclic (i.e. there are no directed loops within the causal graph, meaning it can be expressed as a Directed Acyclic Graph).

Linearity: The relationships between observed variables are linear. Note that this assumption may likely be violated in real-world data but despite this, linear methods can produce much better results when analysing simple causal relationships since they require a much smaller sample size than non-linear methods to be effective.

Continuity: The observed random variables are continuous.

Independence of exogenous variables: This implies that there are no latent confounders affecting more than one observed variable.

This allows us to construct the following model [1]:

$$x_i = \sum_{j:k(j)<k(i)} b_{ij} x_j + e_i$$

where:

- x_i are the observed variables which, without loss of generality, have zero mean
- e_i are continuous, exogenous latent variables that are assumed to be independent of each other, have zero mean and have non-zero variance. This independence assumption implies that there are no latent confounders in the model.
- b_{ij} are the connection strength coefficients from x_j to x_i
- $k(i)$ are the causal orderings detailed earlier, allowing the causal relations of the variables x_i to be represented by a DAG

In matrix form, this gives:

$$\mathbf{x} = \mathbf{B}\mathbf{x} + \mathbf{e}$$

where:

- $\mathbf{x} \in \mathbb{R}^n$ is the vector of observed variables
- $\mathbf{B} \in \mathbb{R}^{n \times n}$ is the connection strength matrix
- $\mathbf{e} \in \mathbb{R}^n$ is the vector of the respective exogenous variables

Note that row and column operations can be applied to **B** to make it strictly lower triangular [1]. Additionally, all non-zero entries imply the existence of a causal relationship, giving an upper bound of $n(n-1)/2$ causal relationships held in a strictly lower triangular matrix.

A simple example is as follows:

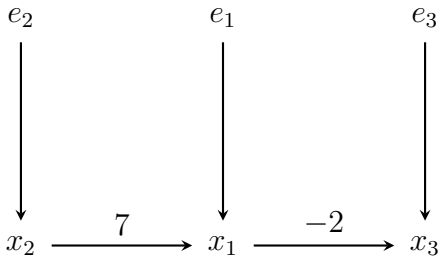


Figure 3 Example of a causal graph representing a linear acyclic SEM with no latent confounders

The SEM corresponding to the example graph in Figure 3 can be written as:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & 7 & 0 \\ 0 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}.$$

Then, permuting the observed variables according to the causal ordering $k(1) = 2$, $k(2) = 1$, $k(3) = 3$, we obtain a strictly lower triangular connection strength matrix:

$$\begin{bmatrix} x_2 \\ x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 7 & 0 & 0 \\ 0 & -2 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ x_1 \\ x_3 \end{bmatrix} + \begin{bmatrix} e_2 \\ e_1 \\ e_3 \end{bmatrix}.$$

Note that the causal ordering required to make a strictly lower triangular connection strength matrix is not necessarily unique [1] (though it is in this case).

Our goal is to estimate the unknown connection strength matrix, **B**, using only the data, **x**, under the assumption that **x** is randomly sampled from a linear acyclic SEM with no latent confounders, similarly to the example above.

B is called identifiable if and only if it can be uniquely estimated or determined from the data distribution of **x** [1]. Identifying the true model will give us the causal structure, and together with the distribution of the exogenous variables, the distribution of the observed variables.

3.2 The Causal Markov Condition and Faithfulness Assumption

The causal Markov condition is a principle that states that each observed variable x_i is independent of its non-descendents in the DAG conditional on its parents, $pa(x_i)$ [1]. In other words:

$$Pr(\mathbf{x}) = \prod_{i=1}^{d} Pr(x_i \mid pa(x_i))$$

The condition holds for any linear acyclic SEM [1] and is very useful for estimating causal structures.

The faithfulness or stability assumption states that the conditional independence of x_i is represented by the graph structure only and not by the specific values of b_{ij} [1]. This means that certain special cases are excluded as they produce invalid SEMs. An example of an SEM and corresponding causal graph where the faithfulness assumption is violated is below:

$$\begin{aligned} x_1 &= e_1 \\ x_2 &= x_1 + e_2 \\ x_3 &= x_1 - x_2 + e_3 \end{aligned}$$

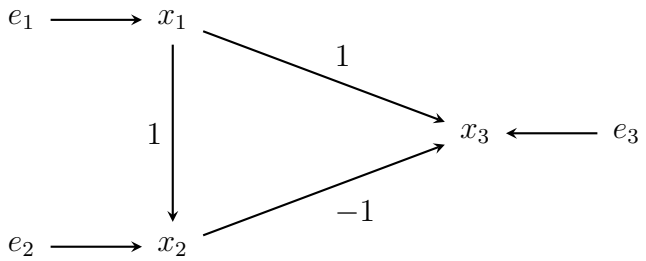


Figure 4 Example causal graph where the faithfulness assumption is violated

where e_1, e_2 and e_3 are mutually independent. Applying the Markov condition to the causal graph, there should be no conditional independence of x_i should hold yet we can see from the equations that x_1 and x_3 are independent.

In practice, especially when working with large datasets, the chance of such an exact cancellation occurring is very low since the set of cases where this happens has measure zero. Hence, the faithfulness assumption is not normally problematic.

However, problems can occur when relying on the Causal Markov condition and faithfulness assumption to estimate the causal structure. In the example below, the causal structure cannot be uniquely identified using these principles alone:

3.2.1 Example of the insufficiency of the Causal Markov Condition

Consider some data generated from the SEM in Figure 3, reproduced in a more simplified form below:

$$x_2 \longrightarrow x_1 \longrightarrow x_3$$

According to the Causal Markov condition, we can conclude from this graph that x_2 is independent of x_3 conditional on x_1. And this is the only conditional independence that holds and so all the independence that we can discover using various methods.

However, there are 3 different causal structures within the class of linear acyclic SEMs that show this same conditional independence which are given in the diagram below. We are unable to determine which is the true structure of the data using the Causal Markov condition and faithfulness assumption alone.

$$x_2 \longrightarrow x_1 \longrightarrow x_3$$

$$x_2 \longleftarrow x_1 \longrightarrow x_3$$

$$x_2 \longleftarrow x_1 \longleftarrow x_3$$

Figure 5 Possible causal structures with the same conditional independence

3.3 The usefulness of Non-Gaussianity

A problem that can occur when using an estimation algorithm to find the underlying SEM for observed data is that there are often multiple different linear acyclic SEMs with no latent confounders with the same conditional independence that equally fit the data, as we have seen in the example above.

If Gaussianity of the exogenous variables is assumed, more than one linear acyclic SEM can show the exact same Gaussian distribution and so equally fit the data, like in the example below, reproduced from [1]:

3.3.1 Example of the problem of Gaussianity [1]

Consider the two simple SEMs below with opposing causal directions:

$$\text{Model 1}: \begin{cases} x_1 = e_1 \\ x_2 = 0.8x_1 + e_2 \end{cases} \qquad \text{Model 2}: \begin{cases} x_1 = 0.8x_2 + e_1 \\ x_2 = e_2 \end{cases}$$

Where in Model 1, $e_1 \sim N(0,1)$ and $e_2 \sim N(0, 0.6^2)$ are independent exogenous variables.

And in Model 2, $e_1 \sim N(0, 0.6^2)$ and $e_2 \sim N(0,1)$ are independent exogenous variables.

So that in both models, $x_1, x_2 \sim N(0,1)$ have the same distribution as shown in Figure 6 below. This is a problem as we are unable to distinguish between the different models using their distributions.

In matrix form, the two models can be written as:

$$\text{Model 1}: \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\boldsymbol{x}} = \underbrace{\begin{bmatrix} 0 & 0 \\ 0.8 & 0 \end{bmatrix}}_{\boldsymbol{B}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\boldsymbol{x}} + \underbrace{\begin{bmatrix} e_1 \\ e_2 \end{bmatrix}}_{\boldsymbol{e}},$$

$$\text{Model 2}: \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\boldsymbol{x}} = \underbrace{\begin{bmatrix} 0 & 0.8 \\ 0 & 0 \end{bmatrix}}_{\boldsymbol{B}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\boldsymbol{x}} + \underbrace{\begin{bmatrix} e_1 \\ e_2 \end{bmatrix}}_{\boldsymbol{e}}.$$

Clearly, the connection strength matrices, **B**, in each of the models are completely different.

We can see that in both models $cov(x_1, x_2) = 0.8 \neq 0$, meaning there is no conditional independence present, so we also cannot distinguish between the models using the Causal Markov condition. This means that under the assumption of Gaussianity of exogenous variables, the matrix **B** is not identifiable.

3.3.2 Assuming Non-Gaussianity

It is possible to distinguish between the two models if we instead assume that the exogenous variables, e_1 and e_2 do not follow a Gaussian distribution in each model [1].

If for example, we assume that the exogenous variables are uniformly distributed, the two models now have different distributions as shown in the right column of Figure 6 below. This observation is true for any Non-Gaussian distribution of exogenous variables [1], and so can be used to distinguish between the models and identify **B**.

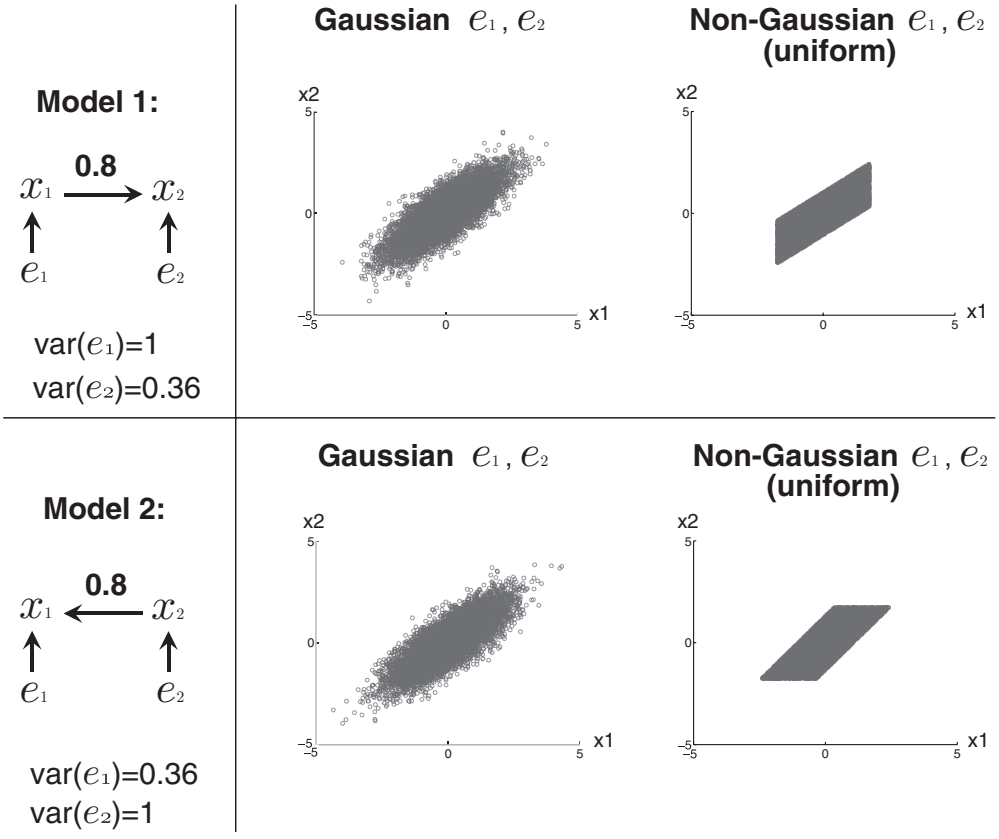


Figure 6 Demonstration of the usefulness of non-gaussianity, reproduced from [1]

In the next few sections we will discuss the methods we used to attempt to identify the causal relationships within a gene dataset, operating under this new, useful assumption of non-Gaussianity.

4 LiNGAM

LiNGAM refers to the Linear Non-Gaussian Acyclic Model and is a non-Gaussian version of the linear acyclic SEM detailed in the following sections:

LiNGAM is defined as:

$$x_i = \sum_{j:k(j)<k(i)} b_{ij} x_j + e_i$$

just like the simple linear acyclic model in the previous section except with the additional assumption that e_i follow a Non-Gaussian distribution

This again gives the matrix form:

$$\mathbf{x} = \mathbf{B}\mathbf{x} + \mathbf{e}$$

LiNGAM has been proven to be identifiable [1] and will form the basis of the rest of this project. We will be using a LiNGAM algorithm to attempt to identify causal structures within real-world genetic data.

4.1 Independent Component Analysis

Independent Component Analysis (ICA) is a non-Gaussian factor of analysis of the form [1]:

$$x_i = \sum_{j=1}^{d} a_{ij} s_j,$$

where x_i are the observed variables and s_j are the known mutually independent continuous latent variables, known as the independent components of the model. The ICA model is a data-generating process, where the latent independent components s_j are summed with the coefficients a_{ij} and are observed as x_i. In matrix form, the ICA model can be represented as [1]:

$$\mathbf{x} = \mathbf{A}\mathbf{s},$$

where **A** is known as the mixing matrix.

The first estimation algorithm of LiNGAM is to transform the model into a ICA model [1], Rearranging LiNGAM, we obatin:

$$\mathbf{x} = (\mathbf{I} - \mathbf{B})^{-1}\mathbf{e},$$

Defining $\mathbf{A} = (\mathbf{I} - \mathbf{B})^{-1}$ we can rewrite the equation as:

$$\mathbf{x} = \mathbf{A}\mathbf{e}.$$

In the context of LiNGAM, the independent components in ICA correspond to the exogenous variables **e**, while the matrix **A** acts as the mixing matrix. ICA states that the mixing matrix **A** can be identified up to permutation, scaling and sign indeterminacies[1]. However, under the assumptions of LiNGAM the mixing matrix **A** is known to be identifiable [1].

Let $\mathbf{W} = \mathbf{A}^{-1}$ and $\mathbf{W}_{ICA} = \mathbf{PDW}$ where **P** is an unknown permutation matrix and **D** is an unknown diagonal matrix. The correct permutation matrix **P** is the only one that contains no zero diagonal elements in **DW**. Since **B** should be a matrix that can be computed to become lower triangular with all zeros and $\mathbf{W} = \mathbf{I} - \mathbf{B}$, the corresponding matrix W must have all

diagonal elements equal to unity. Furthermore, the correct scaling and signs of the independent components can be determined by using the unity on the diagonal of $\mathbf{W} = \mathbf{I} - \mathbf{B}$. To obtain **W**, it is only necessary to divide the rows of **DW** by its corresponding diagonal elements. Finally, the connection strength matrix **B** can be determined from

$$\mathbf{B} = \mathbf{I} - \mathbf{W}.$$

Thus the causal structure (connection strength matrix **B**) is determined from the observational data using ICA.

Limitation of ICA-LiNGAM

ICA-LiNGAM is computationally efficient and asymptotically consistent. However in practice, The stability of ICA-LiNGAM relies heavily on the initial selection of the mixing matrix **A**, since if the initial selection is not properly chosen then the estimation is likely to converge to a local optima (i.e. may not converge to the correct solution). The appropriate selection of such algorithmic parameters can therefore be a complex task.

4.2 DirectLiNGAM

Although the original LiNGAM algorithm estimates the causal structure through Independent Component Analysis (ICA), modern implementations typically use **DirectLiNGAM** [2] since it is guaranteed to converge in a fixed number of steps (equals to the number of variables) when the model assumptions are met and sample size is infinite. Rather than determining the causal structure indirectly through estimation of the mixing matrix, **DirectLiNGAM** estimates the causal ordering of variables directly from the observed data.

The objective of **DirectLiNGAM** is to estimate the connection strength matrix **B** in the LiNGAM model

$$\mathbf{x} = \mathbf{B}\mathbf{x} + \mathbf{e},$$

The **DirectLiNGAM** algorithm proceeds by identifying an exogenous variable, removing its effect from the remaining variables, and repeating the process until a complete causal structure is obtained.

Finding exogenous variable

DirectLiNGAM uses the lemma [2] that, under the LiNGAM assumptions, a variable is exogenous if and only if it is independent of the residuals obtained when the remaining variables are regressed on it. Therefore, for each candidate exogenous variable x_i, the algorithm performs regressions on every other variable x_j ($i \neq j$):

$$x_i = \beta_{ij} x_j + r_i^{(j)},$$

where $r_i^{(j)}$ denotes the residual, and β_{ij} denotes the regression coeffcient.

DirectLiNGAM then searches for the variable $\tilde{x}_m$ that exhibits the greatest degree of independence from its associated residuals given by [2]:

$$\tilde{x}_m = \arg\min_{j \in U \setminus K} \sum_{\substack{i \in U \setminus K \\ i \neq j}} \left(\left| \mathrm{corr}\left(g(\tilde{x}_j),\, r_i^{(j)} \right) \right| + \left| \mathrm{corr}\left(\tilde{x}_j,\, g(r_i^{(j)}) \right) \right| \right).$$

Where $g(\cdot)$ is a nonlinear function (usually chosen as $g(x) = \tanh(x)$ [2]). We therefore select $\tilde{x}_m$ as the exogenous variable and remove its effect from the causal structure.

Residual

Suppose that $\tilde{x}_m$ is identified as an exogenous variable. Its effect is then removed from every remaining variable by linear regression. For each x_i ($i \neq m$), the residual is

$$r_i = x_i - \hat{\beta}_{im}\tilde{x}_m.$$

where $\hat{\beta}_{im} = \frac{\mathrm{Cov}(x_i,\tilde{x}_m)}{\mathrm{Var}(\tilde{x}_m)}$ is the OLS regression coefficient obtained by regressing x_i on $\tilde{x}_m$. DirectLiNGAM uses another lemma [2] that the residual vector

$$\mathbf{r} = (r_1, \ldots, r_{p-1})^T$$

follows another LiNGAM model. Therefore the same exogenous variable search can be applied to the residual variables. Thus the procedure can be repeated to obtain a complete causal ordering of all variables.

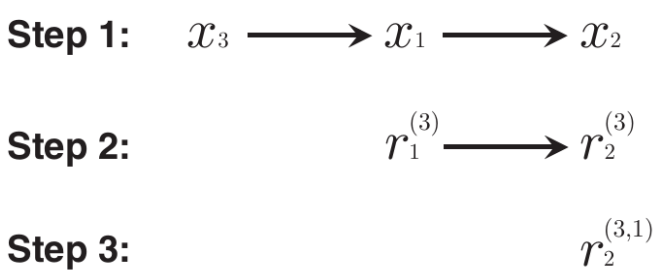


Figure 7 An illustration of DirectLiNGAM: $r_2^{(3,1)}$ denotes the residual when $r_2^{(3)}$ is regressed on $r_1^{(3)}$. [1]

Total causal effect

In addition to direct causal effects, it is often useful to estimate total causal effects. Total causal effect measures the overall effect in one variable resulting by modifying another variable, taking into account both direct and indirect causal pathways. If **B** is the estimated connection strength matrix, the matrix of total causal effects is given by

$$\mathbf{A} = (\mathbf{I} - \mathbf{B})^{-1}.$$

where the entry a_{ij} represents the total causal effect of variable x_j on variable x_i. In the context of gene regulatory networks, total causal effects describe the overall effect of one gene on another through the network. This enables the separation between genes that directly affect a target gene from those who cause an effect through intermediate genes.

5 Data Setup

The following sections make reference to our practical implementation via Python,this can be found on our project [3]

We used the CausalBench [4] dataset, specifically the K562 day 6 Perturb-seq (KD7) dataset, as this would already provide a more than sufficient volume of data for the purpose of the project. This dataset uses K562 immortalised cells, perturbed with CRISPRi targeting DepMap essential genes, with data collected 6 days following perturbation. Essentially, CRISPRi is a method of reducing the expression of an individual gene rather than removing it: a catalytically inactive Cas9 is guided to the target gene's promoter, where it blocks transcription and so lowers — but does not eliminate — that gene's expression. Each cell receives a guide RNA directing this repression at a single target gene, plus a control in which no gene is targeted, and is then profiled by single-cell RNA sequencing. The recorded data is therefore not a count of surviving cells but an expression matrix: for every cell we obtain the measured expression of all genes, tagged with which gene (if any) was repressed in that cell. Repeating this across all the target genes produces the dataset.

We immediately ran into issues with accessing data. The referenced library was only supported on the outdated Python 2, meaning we could not make use of the built in loading functions. This meant we had to first build a pipeline to load directly from the internet, and store the whole dataset locally, the implementation of which is outlined in the Github [3].

Upon loading the data we examined the assumptions of the LiNGAM model, namely acyclicity and non-Gaussianity.

The acyclicity assumption, on the other hand, cannot be checked directly from the data, so we adopt it as a modelling assumption and interrogate it after fitting, through the cyclic verdicts in the Results section. We do still visualise the correlation matrix between genes (Figure 8), but as a way of understanding the data and motivating our choice of method rather than as a test of any assumption; we discuss what it does and does not tell us below.

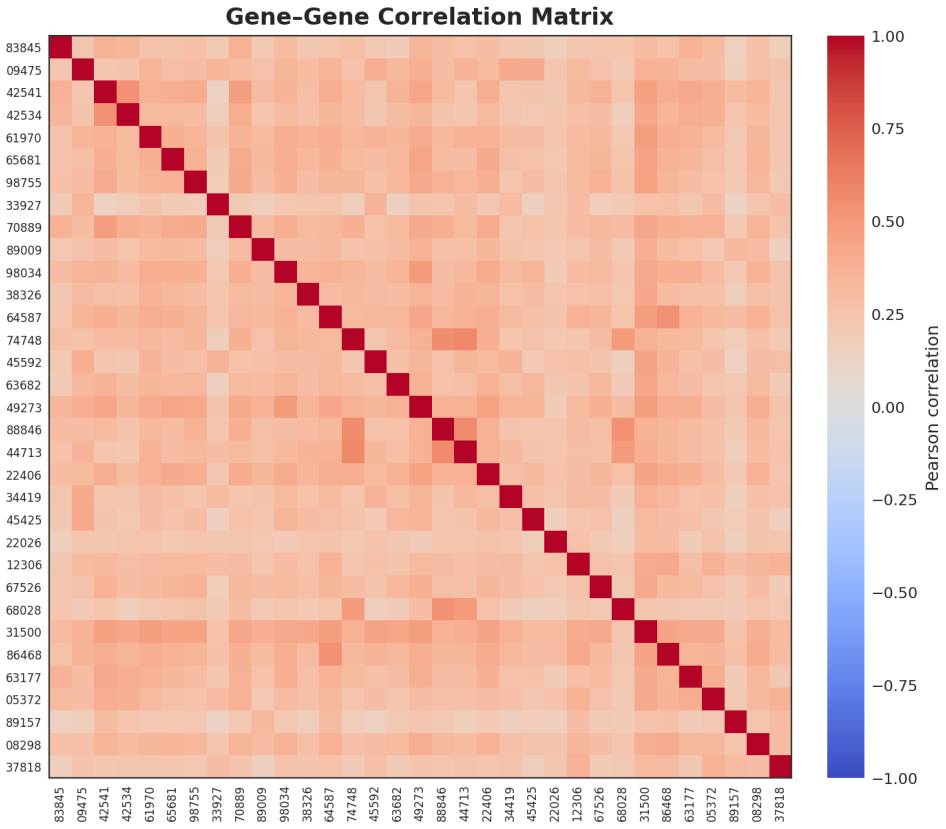


Figure 8 Correlations between the 33 selected genes

Figure 8 shows that the 33 active-everywhere genes used in our fit are almost all moderately correlated: every pair has $|r| > 0.13$, the median is around 0.30, and the most strongly co-varying pair reaches 0.57. We describe in the next subsection how these 33 active-everywhere genes are determined. A correlation matrix is symmetric and so says nothing about the direction in which influence flows; even a purely acyclic chain of genes would produce strong correlations all along it, so this is not in itself evidence of acyclicity.

What the dense correlation does tell us is that a method working from correlations or covariances alone would be of little use here: it would declare almost every gene connected to almost every other, and being symmetric it could not orient any of those links, recovering at best a Markov equivalence class rather than a single directed graph. This is precisely the situation LiNGAM is designed for; it exploits the non-Gaussianity of the structural residuals, checked below, to break that symmetry and recover a unique causal ordering. This assumption will be experimented with when implementing the LiNGAM model.

Next we check the non-Gaussianity assumption. Here, we only consider the genes with the 10 highest variances in expression value, as those would contain the most meaningful biological signal. Out of these genes, we then look at the pair of genes with residuals following distribution closest to Gaussian (smallest absolute excess kurtosis).

5.1 An example use of kurtosis on the data

Kurtosis is the fourth standardised moment of a distribution, namely:

$$\text{Kurt}[X] \equiv \frac{\mu_4}{\sigma_4} = E\left[\left(\frac{X-\mu}{\sigma}\right)^4\right]$$

and is a measure of the tailedness of a distribution.

Excess kurtosis subtracts 3 from the kurtosis of a distribution to calibrate it against a Gaussian, as Gaussian distributions have excess kurtosis 0. This then allows us to take the absolute value of excess kurtosis as a measure of how similar a distribution is to a Gaussian.

The data produces the following plots:

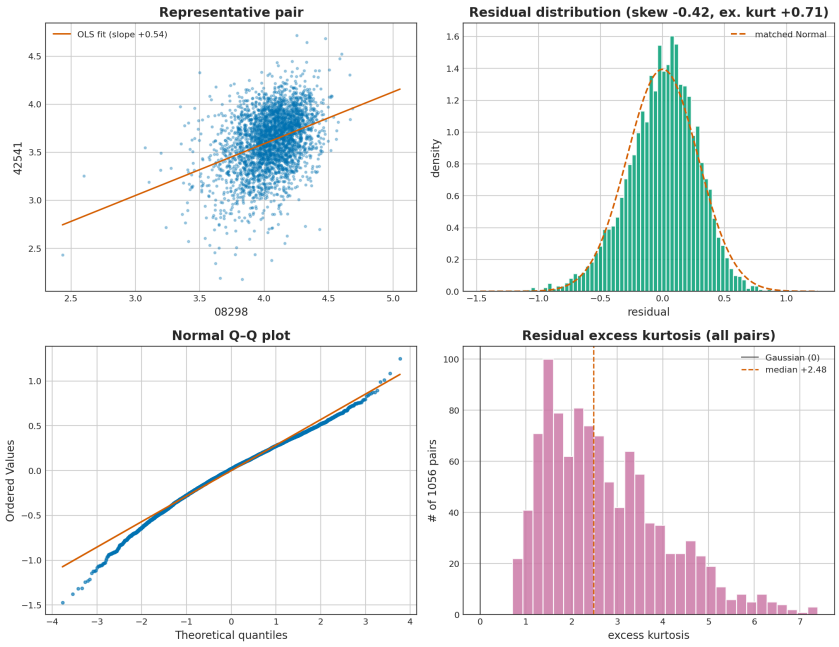


Figure 9 Plots to support non-Gaussian assumption

The upper left plot shows the regression between the pair of genes with most Gaussian residuals, the upper right plot a histogram of the residuals themselves, plotted against a univariate Gaussian, and the lower left a Q-Q plot of these residuals. It is clear from these three plots that the residuals are non-Gaussian, especially from the Q-Q plot where the ordered values deviate from the normal line within the second quantile. The lower right plot shows a histogram of excess kurtosis across all pairwise residuals, and it is clear that they almost all deviate significantly from 0, which combined with the fact that our data has large n, i.e. many samples, indicates that the residuals are likely in fact non-Gaussian, and therefore supports the second assumption necessary for a LiNGAM model.

5.2 Gene selection

Before fitting DirectLiNGAM, a gene-selection step was applied to obtain a suitable working set. Schultheiss and Bühlmann restrict their analysis of the K562 dataset to genes whose expression values are greater than zero in every observational measurement [5]. This reduces the effect of zero-inflation in the single-cell data and gives a set of genes that remain active throughout the observational sample.

We implemented the same active-everywhere filtering principle in the `select_genes` method of `lingam_model.py` [3]. We retained only genes whose `log1p`-normalised expression values were strictly positive in every observational cell. We also required each retained gene to have a corresponding CRISPRi intervention environment, so that inferred outgoing relationships could subsequently be assessed using the interventional data.

The remaining genes were ordered by their observational variance. The implementation allows an optional top-k cap after filtering; however, no additional cap was applied in the final analysis. Starting from 1,158 genes, the procedure retained 33 genes. The final observational expression matrix used for model fitting therefore contained 8,553 cells and 33 genes.

6 Gene Prediction Implementation

6.1 Problem formulation

From the previous section it seems likely that the necessary assumptions for the LiNGAM model are satisfied on our dataset, and therefore it is reasonable to try and make predictions using LiNGAM. Fundamentally, the question we are trying to answer is: **What kind of influence do genes have on one another, and can we create a causal graph to represent these gene-gene relationships.**

Concretely, this means estimating the matrix $\mathbf{B}$ of the LiNGAM model $\mathbf{x} = \mathbf{B}\mathbf{x} + \mathbf{e}$ from the observational expression data alone, where a non-zero entry b_{ij} is read as a direct causal effect of gene x_j on gene x_i. The rest of this section describes how we obtain such a $\mathbf{B}$, how we fit DirectLiNGAM to produce a point estimate, and how we use the bootstrap both to separate stable structure from finite-sample noise and to ask whether a different aggregation of the bootstrap would select a better set of edges. The recovered structure is then validated against the held-out interventional data in the results section.

6.2 Fitting DirectLiNGAM

We fit DirectLiNGAM to the $8{,}553 \times 33$ observational submatrix using the lingam library. To identify exogenous variable, DirectLiNGAM evaluates the degree of independence between the candidate variable x_i and the residuals obtained when the remaining variables are regressed on

it

$$\tilde{x}_m = \arg\min_{j \in U \setminus K} \sum_{\substack{i \in U \setminus K \\ i \neq j}} \left(\left| \mathrm{corr}\Big(g(\tilde{x}_j),\, r_i^{(j)}\Big) \right| + \left| \mathrm{corr}\Big(\tilde{x}_j,\, g(r_i^{(j)})\Big) \right| \right), \qquad g(x) = \tanh(x),$$

introduced in the DirectLiNGAM subsection, $\tilde{x}_m$ would be decided as the exogenous variable by passing the residuals through a $\tanh$ nonlinearity. It works by exploiting the non-Gaussianity of the residuals rather than their covariance, which is exactly the heavy-tailed structure found during our exploration in the last section. The non-Gaussianity is also what makes the recovered ordering unique. A method working from covariances alone could only determine a Markov equivalence class, whereas the non-Gaussian statistic breaks the symmetry and orients every edge [1].

The fit returns a single connection-strength matrix **B**, which we treat as the point estimate of the SEM coefficients. As we detail in the Results section, we treat an edge from x_j to x_i as present when $|b_{ij}| > 0.01$, where the threshold corresponds to "structurally absent" in log1p-normalised expression units. From **B** we also recover the matrix of total causal effects $\mathbf{A} = (\mathbf{I} - \mathbf{B})^{-1}$ introduced earlier, because **B** is acyclic, $(\mathbf{I} - \mathbf{B})$ is non-singular under the recovered ordering and the inverse always exists, and the entry a_{ij} aggregates the effect of x_j on x_i along every directed path between them. We note that because CRISPRi is a soft intervention, knockdown as opposed to knockout, rather than a hard $\mathrm{do}(\cdot)$ operation, **A** should be read as a directionally-correct guideline rather than a precise effect size, something we will consider in the Discussion section.

6.3 Assessing edge stability with the bootstrap

A single point estimate will not help us decide which edges to believe. Since the residuals have heavy tails, just a few outliers among the cells may cause a sign change for marginal regression, adding or removing an edge from our estimate. To separate noise from structure, we evaluate all edges using the bootstrap method.

We create 100 non-parametric bootstrap samples of the 8,553 cells by sampling with replacement. For each sample, we re-run DirectLiNGAM algorithm and generate an adjacency matrix. For each ordered pair (i, j), we end up with a distribution of the coefficient b_{ij}. We get two statistics out of this distribution: selection frequency, defined as the fraction of times $|b_{ij}| > 0.01$, and a median together with inter-quartile range of b_{ij} for the selected values of this coefficient. Only edges with selection frequency of at least 0.80 are considered stable. The most credible claims correspond to edges with a high frequency and tight inter-quartile range. We present the counts of recovered and stable edges in the Results section.

The full procedure is summarised in Algorithm 1. The key design choice, discussed below, is that the edge set $\mathcal{E}$ is fixed once and for all by the point estimate; the loop only attaches a stability annotation to those edges, and never recomputes the (expensive) total-effect matrix.

Algorithm 1: Bootstrap edge-stability annotation

Input: observational matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$; point estimate $\hat{\mathbf{B}}$; number of resamples R; threshold $\tau = 0.01$

Output: selection frequency f_{ij} and median/IQR of b_{ij} for each edge

$\mathcal{E} \leftarrow \{\, (i,j) : |\hat{b}_{ij}| > \tau \,\}$ `// edge set fixed by the point estimate`

for $r \leftarrow 1$ **to** R **do**

  $\mathbf{X}^{(r)} \leftarrow$ sample n rows of $\mathbf{X}$ with replacement

  $\mathbf{B}^{(r)} \leftarrow$ DIRECTLINGAM($\mathbf{X}^{(r)}$) `// no total-effect computation`

  store $b_{ij}^{(r)}$ for all i, j

end

foreach $(i,j) \in \mathcal{E}$ **do**

  $f_{ij} \leftarrow \dfrac{1}{R} \displaystyle\sum_{r=1}^{R} \mathbf{1}\Big[\, |b_{ij}^{(r)}| > \tau \,\Big]$

  record median and IQR of $\{\, b_{ij}^{(r)} : |b_{ij}^{(r)}| > \tau \,\}$

  if $f_{ij} \geq 0.80$ **then** mark (i,j) as stable

end

There are two considerations related to implementation that warrant mentioning, as they can result in errors if overlooked. First, the point estimate **B** determines the edge set, not the bootstrap itself. The latter just adds annotations in the form of a stability frequency for these edges. This is vital since the aggregation of edges between resamples may introduce a false bidirectionality of the graph. Namely, if in one resample a pair is nearly a tie but points towards i, and then switches to j, naive aggregation of the results would give us bidirectional edges although no single model had cycles. Using the bootstrap only as a means of calculating confidence measures will help us avoid this error.

Secondly, we implement our own resampling loop instead of using the library's built-in bootstrap routine. The bootstrap method retrains the model on every sample and also calculates total effects for each possible pair. This step requires counting all possible directed paths, which is $\mathcal{O}(n^3)$ complex according to the number of edges. In the case of a denser network, such as K562, one iteration takes up more than 30 seconds, while the fitting itself takes around 6 seconds. In fact, since we already know $\mathbf{A} = (\mathbf{I} - \mathbf{B})^{-1}$, we only need the per-edge frequency and inter-quartile range. A direct resampling loop makes the 100-resample run manageable.

6.4 Choosing how to aggregate the bootstrap

Maintaining the edge set from the point estimate is a reasonable approach, although it is important to consider whether a different method of aggregation of the bootstrapping results will yield a superior edge set. To answer this question, we compare four selection strategies using one run of the interventional validation described in Results as our ground truth. In doing so, we ensure that each selection strategy will be evaluated against the same held-out evidence, and any differences between the methods will be attributable to the selection strategy, not the validation procedure. An edge $x_j \to x_i$ is considered to be confirmed for the purposes of this investigation if knockdown of the gene x_j yields a Benjamini–Hochberg significant Mann–Whitney shift (described in section 7.3) in gene x_i in the forward direction.

It should be stressed that this criterion is intentionally broader than the stricter confirmed outcome of section 7.3, since in addition to the forward direction being significant, it imposes that the reverse direction be non-significant. In contrast, our criterion involves taking the union of the two criteria, hence the high confirmation rates observed for the table (0.84-0.89) cannot be compared directly to the 10.45% of confirmed edges in our analysis. They answer the weaker

question of whether the predicted direction shows any significant shift.

The four criteria are: (1) *point estimate* baseline criterion, $|b_{ij}| > 0.01$ using the point estimate for the single fit with full data; (2) *edge frequency*, retain only those edges that occur with a selection frequency greater than some threshold value; (3) *median effect*, retain only those edges that have an average coefficient larger than some threshold value among all their occurrences; and (4) *directional frequency*, for each unordered edge pair, retain only the majority edge if its selection frequency dominates some threshold. A representative slice of the comparison is shown in Table 1.

Method	Threshold	Edges	Confirmation rate	Mean W (confirmed)
Point estimate	0.01	335	0.839	0.229
Edge frequency	0.50	285	0.870	0.233
Edge frequency	0.60	203	0.892	0.237
Edge frequency	0.80	85	0.859	0.225
Median effect	0.05	503	0.859	0.232
Directional ratio	0.80	211	0.872	0.221

Table 1 A representative slice of the bootstrap-aggregation comparison (seed $= 0$, 100 resamples, Benjamini–Hochberg $\alpha = 0.05$). Every method is scored against the same fixed interventional ground truth. Higher thresholds trade fewer edges for a higher confirmation rate — the precision/recall knob. The full grid is produced by `bootstrap_experiments.py` [3].

There are two stand out observations here. First, the stability-based filter always increases the precision: using bootstrap frequency or stability filter leads to an increase in the confirmation rate from 0.839 to around 0.87–0.89. The best compromise between the number of edges kept (203 edges) at a 0.892 confirmation rate appears to be edge frequency filter with threshold 0.60. This is an obvious result – thresholding leads to the selection of reproducible edges, and this feature is valued in our interventional dataset. Finally, we reran the baseline experiment using different numbers of resampling operations: 50, 100, and 200, but did not see any change in the confirmation rate at each stability threshold; hence, we could conclude that 100 resamples were sufficient and the result is not an artefact of the bootstrap size.

For the report itself will still use point-estimate edges, however. They are a part of the output from the model, and they ensure recall, on which the false omission calculation in the results section relies – precision filtering that would remove these edges would also remove any that we want to count as missed. We therefore report every point-estimate edge annotated with its bootstrap frequency, and treat the frequency and directional thresholds above as an optional stability filter when a higher-precision sub-graph is wanted.

6.5 A bootstrap diagnostic for feedback

An natural follow-up question then concerns whether the bootstrap test itself could somehow detect feedbacks. For each unordered edge $\{i, j\}$, we compute the number of bootstrap samples that gave an edge $i \to j$ or $j \to i$ and classify $\{i, j\}$ as bootstrap cyclic when both edges occurred in at least a fraction `min_frac` of bootstrap tests. As compared to those edges flagged `cyclic` from the interventional validation test (i.e. significant in both directions), the procedure finds roughly 50 percent, as outlined in Table 2.

min_frac	Pairs flagged	Overlap with interventional	False positives	Recall
0.20	198	128	70	0.520
0.30	115	76	39	0.309
0.40	47	30	17	0.122

Table 2 The bootstrap-cyclic heuristic against the 246 interventionally-cyclic unordered pairs. As the threshold tightens, fewer pairs are flagged and recall of the interventional cyclic set falls.

We emphasize, however, that this is a heuristic and not a test. Inference via DirectLiNGAM occurs under the assumption of acyclicity and always results in an acyclic DAG, so interpreting the switching of a pair across resamples as indicating a cycle represents inference beyond the scope of the procedure. Such switches may result from independence scores being nearly tied, or just from sampling variation due to the heavy tails in this dataset, rather than any form of feedback between the variables involved. The results of our interventional validation in the Results section represent the gold standard for cyclicity, and the bootstrap should be understood as an additional diagnostic identifying pairs of interest. The agreement between the two sets, while not perfect, is nonetheless reassuring and concords with that reported by Schultheiss and Bühlmann for the same K562 dataset [5].

7 Results

7.1 Recovered causal structure

Applying DirectLiNGAM to the filtered observational dataset produced a connection-strength matrix B. An edge from x_j to x_i was treated as present when the corresponding coefficient satisfied

$$|b_{ij}| > 0.01.$$

We discuss why this threshold was used in the next subsection.

The point estimate of the recovered graph contained 335 directed edges.

To assess the stability of these inferred relationships, the model was fitted to 100 bootstrap samples. An edge was classified as stable when it appeared in at least 80% of the bootstrap fits. Of the 335 directed edges in the point estimate, 83 satisfied this stability criterion. Therefore, although the recovered graph contains a relatively large number of proposed relationships, a smaller subset is consistently recovered under resampling.

Figure 10 shows a readable 15-node subset of the recovered causal graph. The complete 33-node graph is not displayed, since its density would make the individual relationships difficult to interpret clearly. In Figure 10, the width of an edge is proportional to the magnitude of its estimated coefficient, while its opacity reflects its bootstrap selection frequency.

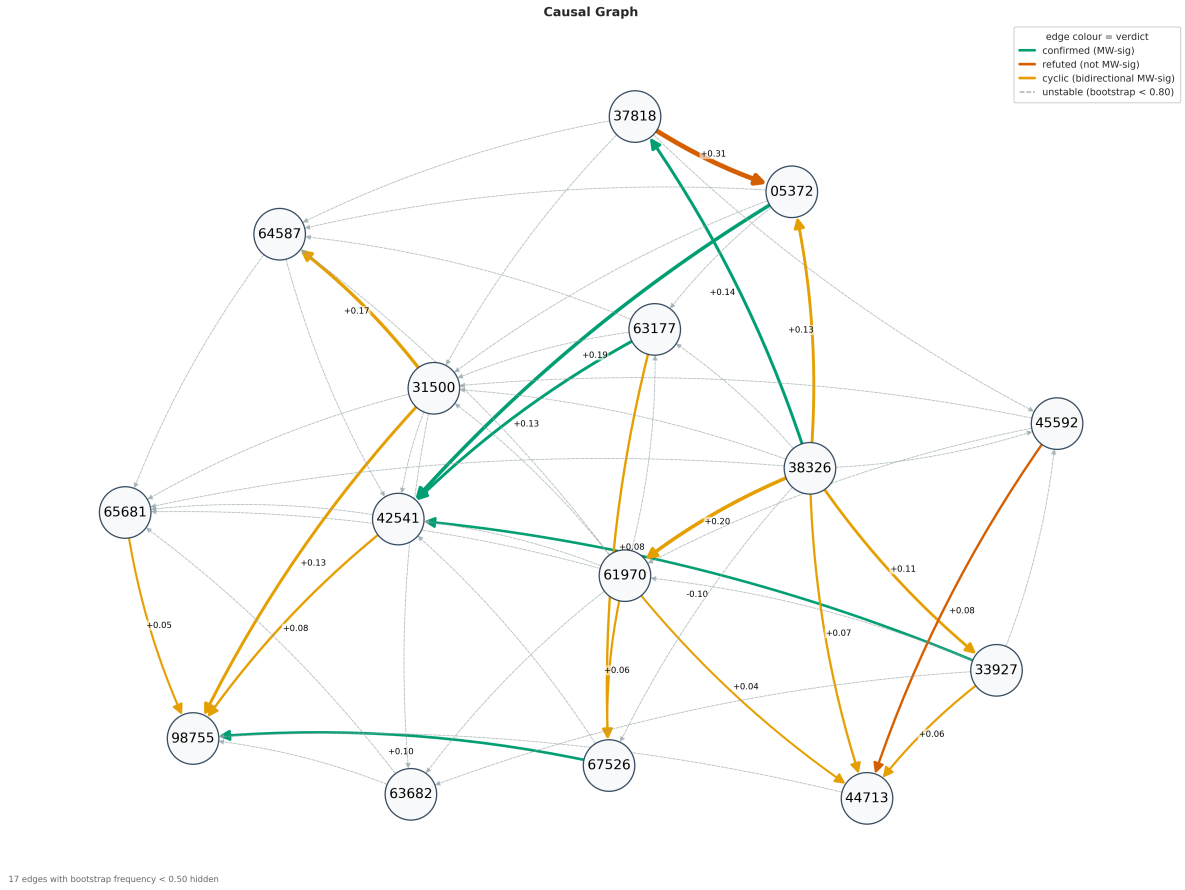


Figure 10 A 15-node subset of the causal graph recovered by DirectLiNGAM from the observational K562 data. Edge width is proportional to the magnitude of the estimated coefficient and opacity reflects bootstrap selection frequency. Stable edges are coloured according to their interventional validation outcome: green edges are confirmed, red edges are refuted and orange edges are classified as `cyclic`. Edges below the bootstrap-stability threshold are shown in grey.

7.2 Sensitivity to the edge threshold

The cutoff $|b_{ij}| > 0.01$ is inherited as the lingam library default rather than tuned, so we re-ran the interventional validation at $\tau \in \{0.005, 0.01, 0.02, 0.05\}$, reusing the single point estimate $\mathbf{B}$ (which does not depend on τ) and varying only the edge/no-edge split. The results are shown in Table 3. In essence, the verdicts are almost flat across 0.005–0.02: the fit produces no coefficients in the interval $(0.005, 0.01]$, so the default sits on a plateau rather than a cliff, and the recovered edge count, the confirmed rate ($\approx 10.4\%$) and the false-omission rate ($\approx 86.6\%$) are all unchanged. The structure changes only at $\tau = 0.05$, with the number of edges reducing from 333 to 221 due to the weakest edges becoming primarily indirect links; still, the confirmed rate (12.2%) and the false omission (87.0%) change only slightly at that threshold. Thus we choose to keep the default value of 0.01. It is in the middle of a stable range, and because the false-omission analysis requires all point-estimation edges to be maintained, there is no reason to increase the threshold to gain the insignificant increase in precision observed at 0.05.

Threshold	Predicted edges	Confirmed	Confirmed rate	Mean W (pred.)	FOR
0.005	335	35	10.4%	0.199	86.6%
0.010	335	35	10.4%	0.199	86.6%
0.020	333	35	10.5%	0.200	86.6%
0.050	221	27	12.2%	0.203	87.0%

Table 3 Sensitivity of the interventional verdicts to the edge-presence threshold $\tau = |b_{ij}|$. The point estimate $\mathbf{B}$ is fitted once; only the edge/no-edge split and path reachability change with τ. FOR is the false-omission rate over no-path pairs.

7.3 Interventional validation

The relationships recovered from the observational data were compared with the separately retained CRISPRi intervention environments. Following the evaluation framework used in CausalBench [6], we compared the expression distribution of a proposed effect gene after suppression of the proposed cause gene with its distribution in non-targeting control cells.

The size of the difference between the intervention and control distributions was measured using the Wasserstein distance. This measures how far one probability distribution must be shifted to match another: a larger value therefore indicates a greater change in the expression distribution following intervention. Formally,

$$W_1(\mu, \nu) = \int_0^1 \left| F_\mu^{-1}(t) - F_\nu^{-1}(t) \right| \, dt$$

where F_μ^{-1} and F_ν^{-1} are the quantile (inverse CDF) functions of the intervention and control distributions respectively.

Statistical significance was assessed using a two-sided Mann–Whitney U test. This is a non-parametric, rank-based test used here to compare the intervention and control samples. A two-sided test was used because suppression of the proposed cause gene may either increase or decrease the expression of the proposed effect gene. The test statistic is defined as

$$U = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \mathbf{1}[X_i > Y_j]$$

where X_i are the n_1 intervention samples, Y_j are the n_2 control samples, and $\mathbf{1}[\cdot]$ is the indicator function. Since the test was applied repeatedly across many ordered gene pairs, the resulting p-values were adjusted using the Benjamini–Hochberg procedure. An adjusted p value below 0.05 was treated as statistically significant.

The final analysis contains 33 retained genes. Excluding pairs of the form (i, i), there are therefore

$$33 \times 32 = 1056$$

ordered gene pairs. Every one of the $1{,}056$ tuples is directed to exactly one of six possible conclusions by the decision-making algorithm presented in Fig. 11. In essence, the algorithm first checks whether the graph predicts a direct connection between nodes and then checks the interventional data in the direction of interest. The six leaves of the tree are mutually exclusive and collectively exhaustive, which is why the numbers of each verdict in Table 4 add up to precisely $1{,}056$. The verdicts on the right side relate to the 335 tuples where there was a direct prediction, while those on the left side relate to the other 721 tuples.

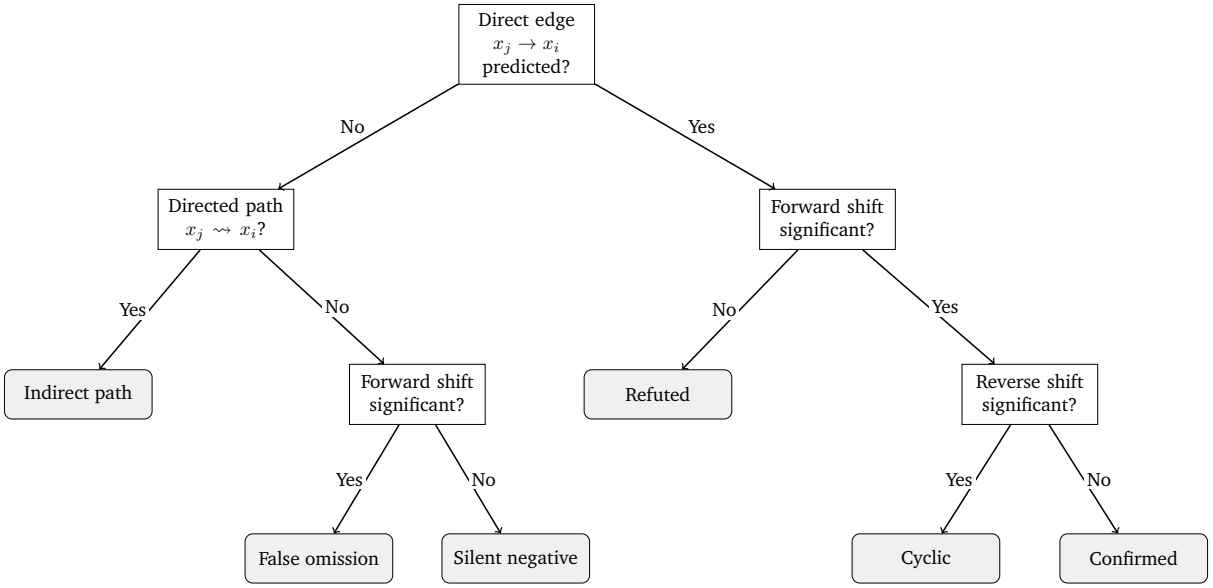


Figure 11 Decision procedure assigning each ordered pair (x_j, x_i) to exactly one of the six verdicts. "Forward shift" refers to the change in x_i under knockdown of x_j; "reverse shift" to the change in x_j under knockdown of x_i. Significance is the Benjamini–Hochberg-adjusted Mann–Whitney verdict at $\alpha = 0.05$.

These verdicts are summarised in Table 4.

Verdict	Number of ordered pairs	Mean Wasserstein distance
Confirmed	35	0.213
Refuted	54	0.044
Cyclic	246	0.231
False omission	471	0.219
Silent negative	73	0.040
Indirect path	177	0.176
Total	1056	

Table 4 Verdict categories for the 1,056 ordered gene pairs. The Wasserstein distance measures the difference between the empirical expression distributions under intervention and control conditions.

A predicted edge was classified as *confirmed* when suppression of the proposed cause gene was associated with a statistically significant shift in the proposed effect gene, while the corresponding reverse-direction comparison was not significant. There were 35 confirmed edges. The mean Wasserstein distance within this category was 0.213, indicating a moderate distributional shift on average.

A predicted edge was classified as *refuted* when suppression of the proposed cause gene did not produce a statistically significant shift in the proposed effect gene. There were 54 refuted edges. The mean Wasserstein distance in this category was 0.044, which is substantially lower than the corresponding value for confirmed edges.

The remaining 246 predicted edges were classified as `cyclic`. An edge received this label when statistically significant shifts were detected in both directions. In other words, suppression of x_j shifted the distribution of x_i, while suppression of x_i also shifted the distribution of x_j. The mean Wasserstein distance for this category was 0.231.

The 335 predicted edges therefore consist of

$$35 + 54 + 246 = 335$$

ordered pairs. The confirmed-edge rate among the predicted edges was

$$\frac{35}{335} = 10.45\%.$$

Figure 12 plots the Wasserstein distance against bootstrap selection frequency for each predicted edge. The dashed vertical line marks the stability threshold of 0.8.

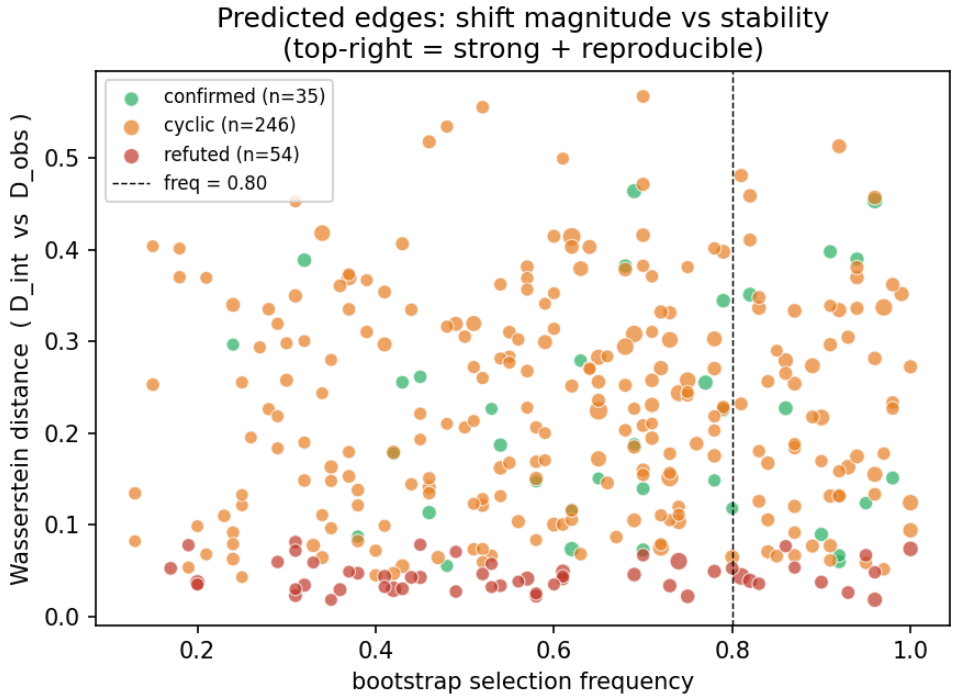


Figure 12 Wasserstein distance between the interventional and control distributions plotted against bootstrap selection frequency for each predicted edge. The dashed vertical line marks the bootstrap-stability threshold of 0.8. Green points are confirmed edges, red points are refuted edges and orange points are classified as `cyclic`.

The scatter plot shows that bootstrap stability and interventional validation measure different properties. Among the 83 stable edges, 11 were confirmed, 12 were refuted and 60 were classified as `cyclic`. Thus, an edge may be recovered consistently from the observational sample without receiving an asymmetric interventional verdict.

7.4 Relationships without a predicted direct edge

The remaining three categories concern ordered pairs for which DirectLiNGAM did not predict a direct edge.

An ordered pair was classified as an *indirect path* when the recovered graph did not contain a direct edge from x_j to x_i, but did contain a directed path through one or more intermediate genes:

$$x_j \longrightarrow \cdots \longrightarrow x_i.$$

There were 177 ordered pairs in this category. This category allows for an intervention effect to propagate through intermediate genes. Its mean Wasserstein distance was 0.176, which is consistent with a moderate distributional shift on average.

For the remaining ordered pairs, the recovered graph contained no directed path of any length. These pairs were divided into two categories. A *false omission* occurred when the model predicted no path from x_j to x_i, but suppression of x_j nevertheless produced a statistically significant shift in x_i. There were 471 false omissions, with a mean Wasserstein distance of 0.219.

A *silent negative* occurred when the model predicted no path and suppression of x_j did not produce a statistically significant shift in x_i. There were 73 silent negatives, with a mean Wasserstein distance of 0.040.

Among the 544 ordered pairs for which the recovered graph predicted no directed path, the false-omission rate was therefore

$$\frac{471}{544} = 86.58\%.$$

The high false-omission rate indicates that the fitted graph fails to represent many relationships that are detectable under intervention.

8 Discussion

8.1 Interpretation of the validation results

The recovered graph provides a partial description of the relationships visible under intervention. The 35 confirmed edges are the clearest outcomes, since suppression of the proposed cause gene produces a statistically significant shift in the proposed effect gene, while the reverse-direction comparison is not significant.

The largest predicted-edge category consists of the 246 edges classified as cyclic. This means that significant shifts were detected in both directions. However, this does not prove that each pair forms a direct biological feedback loop. Other possible explanations include indirect pathways, omitted variables or limitations of the linear acyclic model. Schultheiss and Bühlmann identify similar cyclicity flags when analysing the same K562 dataset [5].

Figure 12 also shows that bootstrap stability and interventional validation measure different properties. An edge may be recovered consistently from resampled observational data without receiving a clear asymmetric interventional verdict. The false-omission rate of 86.58% provides further evidence that the fitted graph does not represent all relationships detectable under intervention.

8.2 Direct and indirect effects

The coefficients b_{ij} represent proposed direct effects. However, suppression of one gene may affect another through intermediate genes, for example:

$$x_j \longrightarrow x_k \longrightarrow x_i.$$

In this case, an intervention on x_j may shift the distribution of x_i even when the recovered graph does not contain a direct edge from x_j to x_i.

This distinction is important when interpreting both the 177 indirect-path pairs and the false omissions. A statistically significant distributional shift does not necessarily validate a single direct edge in isolation. CausalBench also highlights the difficulty of distinguishing direct causal edges from broader causal paths when evaluating network-inference methods on real single-cell data [6].

8.3 Limitations and possible extensions

The interpretation of the recovered graph depends on the assumptions of LiNGAM. In particular, the model assumes linear relationships, acyclicity and mutually independent non-Gaussian error terms [1]. The main limitations are summarised in Table 5.

Assumption	Possible limitation	Effect on interpretation
Linearity	Relationships between genes may be nonlinear.	The estimated coefficients may provide only an approximate description of the effect strength.
Acyclicity	Biological systems may contain feedback mechanisms.	Bidirectionally significant relationships may not be represented adequately by a DAG.
No latent confounders	Genes omitted from the filtered dataset may affect several retained genes.	Some recovered edges may reflect hidden common causes rather than direct effects.
Ideal interventions	CRISPRi suppresses gene activity rather than fixing it at an exact value.	The intervention data do not correspond exactly to the theoretical do-operation.

Table 5 Limitations of the LiNGAM assumptions when applied to the K562 gene-expression data.

Schultheiss and Bühlmann also note that some knockdown observations may lie outside the support of the observational data [5]. The Wasserstein distances should therefore be interpreted as descriptive measures of distributional shift rather than calibrated causal-effect sizes.

The recovered graph remains useful as an exploratory result. It identifies candidate relationships for further investigation and shows how observational causal discovery can be assessed using separately retained interventional data. Future work could compare DirectLiNGAM with methods that allow nonlinear relationships or latent confounding, and assess the sensitivity of the results to the chosen thresholds.

9 Use of AI

The primary usage of AI was for assistance with coding, we used Claude Code built into Visual Studio Code for this. Claude Code was used with aiding the production of the markdown files, in particular for structuring them, the production of the data pipeline also made use of AI, due to our lack of experience with this type of task, its use here also did not in any way impact the actual output of our models. Finally, Claude Code was used to summarise and push when large changes were made.

Additionally we used AI to aid with the formatting of the Latex file.

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